

Introduction

This document is a supplementary document for the paper entitled *Prioritising Pharmaceuticals by Comparing Fish Plasma Concentrations Predicted by a PBK Model with Human Therapeutic Plasma Concentrations* and is designed to guide the user on how to use the R-based PBK model. The files required to run the model can be found at <https://github.com/FunkNoodle/FishPBKModel>. While the model runs in R, it is not required for the user to have extensive knowledge of R or scripting. Prior knowledge will help with bug fixing but is not required. The original (equation based) version of the PBK model can be found [here](#).

R Installation

If you do not have R installed on your computer, you will need to download both base R, and R studio. Base R is required for R studio to work:

R Studio: <https://posit.co/download/rstudio-desktop/>

Base R: <https://cran.rstudio.com/>

R studio is used to run the model.

PBK Model Files

There are four files that are required for the model to run. Three .R script files and a CSV file required for the user input data. These files can be downloaded from <https://github.com/FunkNoodle/FishPBKModel>. Below is a brief overview of what each file does and a screenshot to identify them:

PBK Model.R

This is the master script that contains the PBK model. You will not need to open this script, but if you are interested in how it runs, it uses the 'Example' data in the PBKinput file (which is actually the data for carbamazepine. Do not change anything in this script unless you know what you are doing, otherwise the model will not work as intended.

```

### ><(((?> ><(((?> ><(((?> ><(((?> ><(((?> ><(((?> ><(((?> ><(((?> ##
### Input compound name ###

# Example uses Carbamazepine as an input
target_name <- "Example"

### Input parameters ###

PBKinput <- read.csv("PBKinput.csv")

# Find the row with the target compound name
row <- PBKinput[PBKinput$Name == target_name, ]

# Extract values from the row and assign them to variables
if (nrow(row) == 1) {
  Name <- row$Name
  BM <- row$BM
  Sex <- row$Sex
  temp <- row$temp
  pHw <- row$pHw
  pKa <- row$pKa
  logKow <- (row$logKow)
  logKow_ion <- (row$logKow_ion)
  ii <- row$ii
  HL <- row$HL
  C_water <- row$C_water
  Q_ingest <- row$Q_ingest
} else {
  stop("Name not found or multiple entries for the same name.")
}

#Amount of time between timesteps (days)
dt = HL/10000

# Converting logKow to Kow and changing ionic Kow to Kow if it is missing
Kow <- 10^logKow

if(is.na(logKow_ion)) {
  Kow_ion = Kow
} else {
  Kow_ion <- 10^logKow_ion
}

```

Figure 1: A screenshot of the PBK Model.R script

Batch Code.R

This script will run each of the compounds you tell it to in turn. You will need to edit parts of this script if you want to use it. This will be explained later. Using this script is one of the two options available with this model. Using this version is far slower than the parallel version, but will not slow down your computer as much, so you can still use it while the model runs in the background.

This script works by creating duplicates of the PBK model script and adding the input values from PBKinput.csv for each compound. This results in a new script for each compound that contains its name and the values from the CSV. This script then runs each of those scripts in turn and outputs the results into a new CSV for that compound. For the study, there is an extra bit of code at the end which combines the final values from the venous and arterial concentrations for each compound into a new xlsx file named "combined_plasma_results". You may want to use this if your aims are similar to the paper, but it is not required.

```

#           o   o           /AAAAA7
#           '   '           'oo))))))oo,
#           '   '           '))))))))) /{
#           '   '           'o)))))))))={
#           '   '           ')))))\ \)))))={
#           '   '           ')))))\ \)))))={
#           '   '           ')))))\ \)))))={
#           '   '           'oo))))))oo*'

### Master Script ###

# This runs each .R script in turn

starttime <- Sys.time()
print(starttime)

# Load the necessary library
library(dplyr)

# Read the PBKinput dataframe
PBKinput <- read.csv("PBKinput.csv")

# Read the template script
template_script <- readLines("PBK Model v3.R") # Adjust the path to your template

# Iterate over each name in column A
for (name in PBKinput$Name) {
  # Replace the target_name in the template with the current name
  new_script <- gsub("target_name <- \"Example\"", paste("target_name <- '", name, "'", sep = ""), template_script)

  # Create a new filename for the modified script
  new_filename <- paste0(name, ".R") # Save with the name as part of the filename

  # Write the modified script to a new file
  writeLines(new_script, new_filename)
}

```

Figure 2: A screenshot of the Batch Code.R script

Parallel Batch Code.R

This script will run each of the compounds you tell it to in parallel. You will need to edit parts of this script if you want to use it. This will be explained later. Using this script is one of the two options available with this model. Using this version is much faster than the previous version, but it can slow down your computer considerably, depending how many compounds you are running at once and how powerful your computers CPU is. It uses all but one of the CPU cores, so everything else your computer will be doing will likely be assigned to that single core, hence you should not run this code while doing anything else that requires a heavy load on the CPU. Use this version if you are about to go for a break or while you are grabbing a coffee.

This script works by creating duplicates of the PBK model script and adding the input values from PBKinput.csv for each compound. This results in a new script for each compound that contains its name and the values from the CSV. This script then runs each of those scripts in parallel (i.e. all together at the same time) and outputs the results into a new CSV for that compound. For the study, there is an extra bit of code at the end which combines the final values from the venous and arterial concentrations for each compound into a new xlsx file named "combined_plasma_results". You may want to use this if your aims are similar to the paper, but it is not required.

```

#           o  o           /AAAAA7
#           '  '           ,oo))))))oo,
#           '  '           ,))))))))) ,/f ~~~~~
#           '  '           ,o ))))))))=f ~~~~~ speedyfish
#           '  '           > ))))))))=f ~~~~~ Starting!
#           '  '           , ))))))=f ~~~~~
#           '  '           ' ,))))))\ ))))))' \f ~~~~~
#           '  '           'o))))))o*'

# This runs each .R script in parallel
# Faster than the Batch code, but will slow your computer down

### Master Script ###

starttime <- Sys.time()
print(starttime)

# Load necessary libraries
install.packages(c("doParallel", "foreach"))
library(doParallel)
library(foreach)
library(dplyr)

# Read the PBKinput dataframe
PBKinput <- read.csv("PBKinput.csv")

# Read the template script
template_script <- readLines("PBK Model v4.R") # Adjust the path to your template

# Iterate over each name in column A
for (name in PBKinput$Name) {
  # Replace the target_name in the template with the current name
  new_script <- gsub('target_name <- "Example"', paste('target_name <- "', name, '"', sep = '"'), template_script)

  # Create a new filename for the modified script
  new_filename <- paste0(name, ".R") # Save with the name as part of the filename

  # Write the modified script to a new file
  writeLines(new_script, new_filename)
}

```

Figure 3: A screenshot of the Parallel Batch Code.R script

PBKinput.csv

This file is where the user can input their parameters for the model to use. It is recommended to leave the example row as it is for reference. Below is an explanation of each of the column headers:

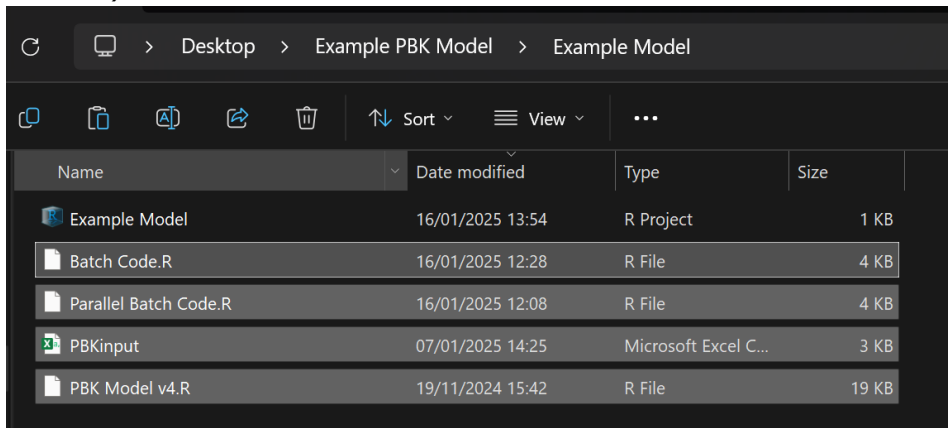
Column Header	Explanation
Name	The name of the compound or analysis. The user is free to input what they like here so long as there are no duplicates. See figure 3 for examples.
BM	Body Mass: The mass of the fish in grams
Sex	The sex of the fish. 1 for male, 2 for female, 0 for mixed/unknown/other
Temp	Temperature of the environment (surrounding water) in Celsius
pHw	The pH of the surrounding water
pKa	The acid dissociation constant of the compound
logKow	The log of the octanol-water partition coefficient (log K _{ow}) of the compound
logKow_ion	The log of the octanol-water partition coefficient (log K _{ow}) of the ionized version of the compound. See the methods section of the main paper to see how to obtain this.
ii	Ionization state: acid = 1, base = -1, neutral = 0
HL	Half-Life: Whole-body primary biotransformation half-life normalised to a 10g fish at 15 °C in days
C_water	The concentration of the compound in the surrounding water (ug/ml or mg/l)
Q_ingest	Exposure concentration through food (ug/d), set to 0 if there is no ingestion

	A	B	C	D	E	F	G	H	I	J	K	L
1	Name	BM	Sex	temp	pHw	pKa	logKow	logKow_ior ii	HL		C_water	Q_ingest
2	01Disulfira	10	0	15	7.8		4.16		0	0.938	0.0000374	0
3	02Felbinac	10	0	15	7.8	4.73	3.258	0.44	1	2.113	0.01133	0
4	03Flecainide	10	0	15	7.8	9.62	3.188	1.24	-1	2.299	0.00684	0
5	04Orphenadrine	10	0	15	7.8	8.87	4.165	0.74	-1	0.1734	0.00303	0
6	05Prazosin	10	0	15	7.8	7.92	1.651	-1.65	-1	0.0009613	0.01411	0
7	06Sulpiride	10	0	15	7.8	8.97	0.125	-2.01	-1	0.003433	0.0014	0
8	07Tribenoside	10	0	15	7.8	12.36	5.17	1.83	1	0.02357	0.01153	0
9	08Bimatoprost	10	0	15	7.8	14.35	2.629	0.45	1	2.948	0.02026	0
10	09Ataluren	10	0	15	7.8	3.9	3.919	0.32	1	3.645	0.02411	0
11	10Procyclidine	10	0	15	7.8	9.58	3.788	1.17	-1	0.1358	0.007	0
12	11Clometholide	10	0	15	7.8	3.18	1.772	-0.91	-1	0.06868	0.05464	0
13	12Nimesulide	10	0	15	7.8	6.4	1.787	-0.17	1	1.985	0.001	0
14	13Tolperisone	10	0	15	7.8	8.74	3.566	0.43	-1	0.1598	0.00597	0
15	14Cilostazol	10	0	15	7.8	14.42	3.306	0.97	1	0.1043	0.01392	0
16	15Mephentermine	10	0	15	7.8	12.68	1.187	-1.89	1	0.02638	0.015	0
17	16Heptamethylnol	10	0	15	7.8	10.43	0.82	-1.43	-1	0.1073	0.174	0
18	17Indometacin	10	0	15	7.8	4.38	3.35	1.27	1	3.912	0.00002	0
19	18Clozapine	10	0	15	7.8	8.16	3.4	0.98	-1	0.8132	0.000179	0
20	19Tolbutamide	10	0	15	7.8	5.64	2.295	-0.44	1	0.02005	0.01566	0
21	20Indapamide	10	0	15	7.8	8.85	2.64	-0.25	1	2.511	0.00621	0
22	21Perazine	10	0	15	7.8	8.68	3.778	1.11	-1	0.01519	0.00314	0
23	22Fluphenazine	10	0	15	7.8	8.61	3.966	1.35	-1	0.01372	0.00667	0
24	23Ketoconazole	10	0	15	7.8	6.42	4.192	0.95	-1	0.9564	0.00014	0
25	24Imiquimod	10	0	15	7.8	5.01	2.651	-0.36	-1	0.7181	0.01102	0
26	25Quetiapine	10	0	15	7.8	7.76	2.808	0.34	-1	0.2301	0.01	0
27	Q1Carbamazepine	10	0	15	7.8	15.96	2.766	0.14	1	0.15	0.002	0
28	Q2Diclofenac	10	0	15	7.8	4.01	4.259	1.85	1	6.13	0.00004	0
29	Q3Diphenhydramine	10	0	15	7.8	8.87	3.652	1.7	-1	0.05	0.00099	0
30	Q4Fluoxetine	10	0	15	7.8	9.4	4.173	1.51	-1	3.89	0.0001	0
31	Q5Ibuprofen	10	0	15	7.8	4.85	3.844	0.75	1	1.88	0.000011	0
32	E1Mycophenolate	10	0	15	7.8	3.57	3.528	-0.08	1	0.01278	0.0058	0
33	E2Olaparib	10	0	15	7.8	9.96	1.964	-0.49	0	0.09475	0.32	0
34	Example	10	0	15	7.8	7.92	1.91	-1.65	-1	0.0009613	0.01411	0

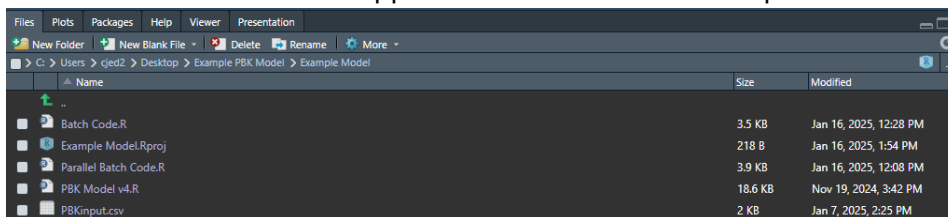
Figure 3: A screenshot of the PBKinput.csv file. The data used in the study is visible here.

Running the model using Batch Code.R

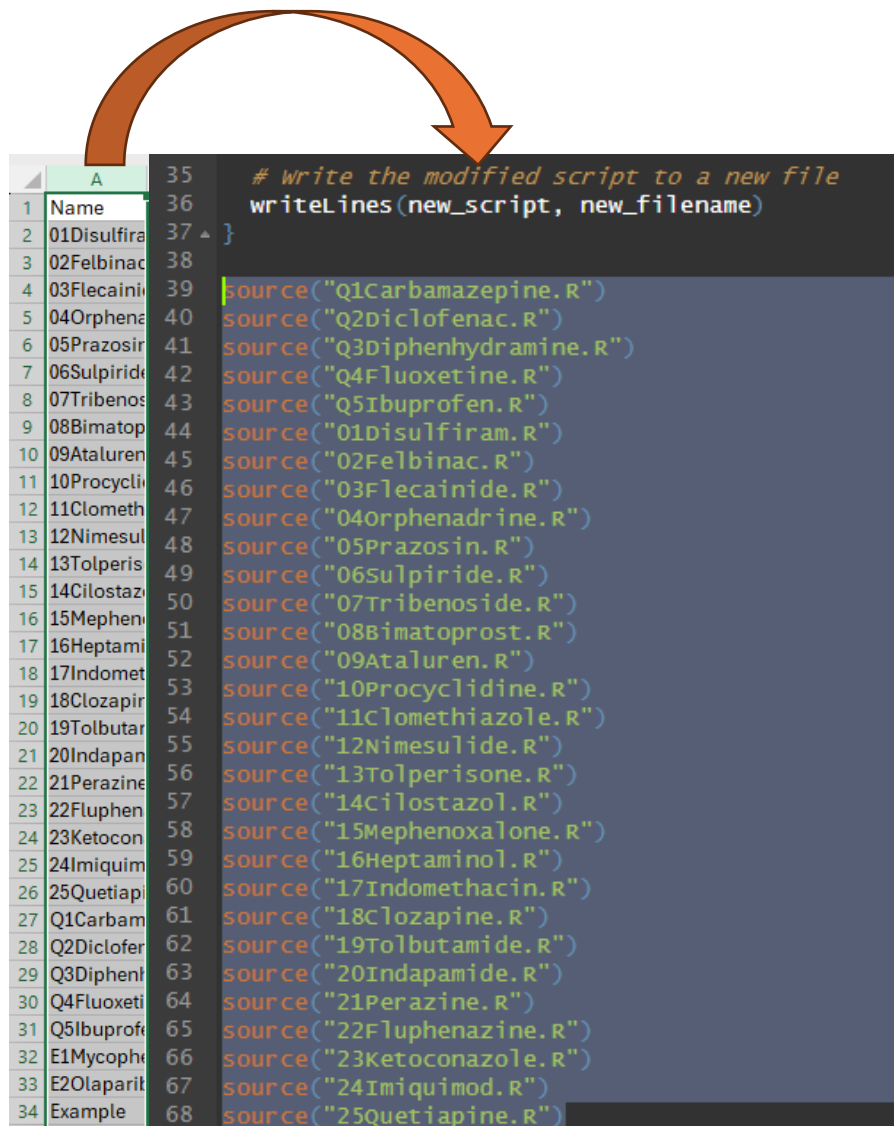
1. File → New Project → New Directory → New Project
 - a. Give your project a name and select an empty folder somewhere on your computer
2. Paste in the model files to the new directory (The three .R files and the PBKinput CSV file):



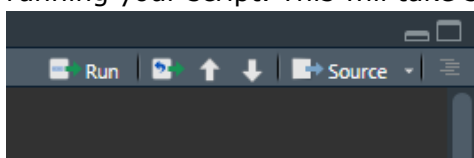
3. These should now appear in the file explorer in R Studio:



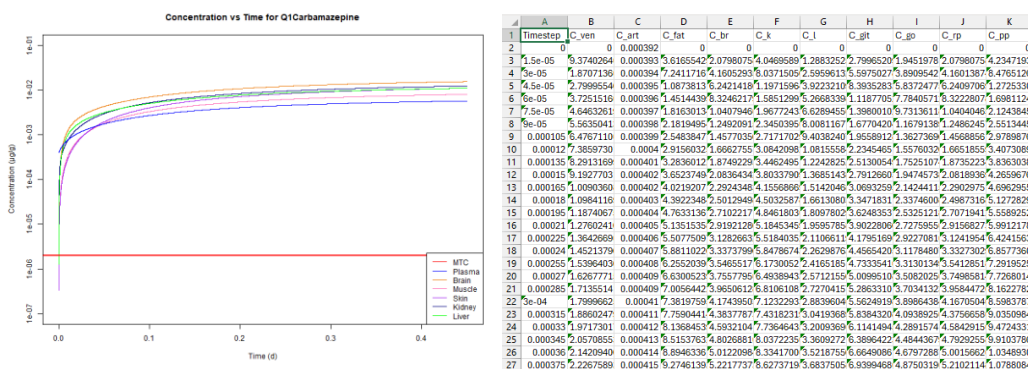
4. Open PBKinput.csv and insert your data. See above for details on each column
5. Open Batch Code.R in R studio. This will open the script. Scroll down to find where the code is going to use the names from the CSV, should be line 39.
 - a. Change the names in green to match the names you put into your CSV file. For example, if you have a row in your Input CSV named "Paracetamol_10mg", then the line in the code should read `source("Paracetamol_10mg.R")`:



6. Press the 'Source' button on the top right of the Batch Code.R script to begin running your script. This will take some time:



7. The model will now go through each compound one by one, and generate a PBKOutputCSV and a graph. You can find both of these either in the folder, or in the file explorer window of R Studio:



- a. The Output CSV contains all of the compartments in the model, and the concentration at each timestep (measured in days). The length of the

timesteps depends on the half-life of the compound ($\text{timestep} = \text{HL}/10,000$).

8. Done!

Once the model has finished running, an ASCII picture of a happy fish will appear in the console, along with the time difference between the start and end of the run, so that you know how long this process takes roughly for future runs.

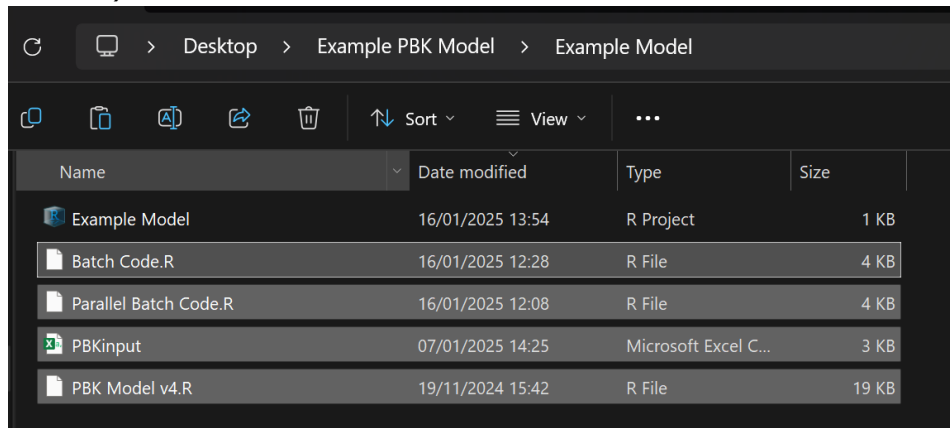
```
> #           o   o
> #           '   '
> #           /AAAAA7
> #           ,oo))))))oo,
> #           ,))))))))) /{
> #           ' ^  ))))))))={
> #           >   ))))))))={
> #           `   )))))\ )))))={
> #           ', ))))))\ /))))' \{
> #           '*o))))))o*'
>
> print(endtime)
[1] "2025-01-16 14:38:26 CET"
> print(endtime-starttime)
Time difference of 32.87615 mins
>
```

Done!

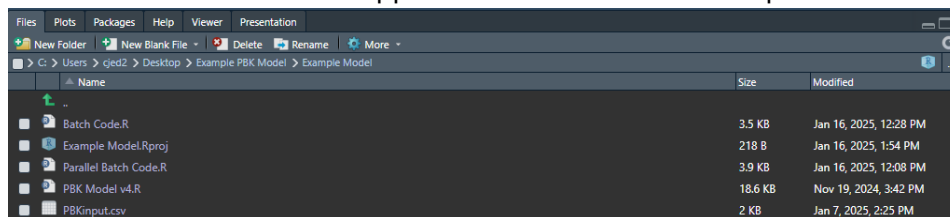
- a. The code also creates an excel file with a combination of all the final blood plasma results. You may also use this if it is useful to you.

Running the model using Parallel Batch Code.R (Faster)

1. File → New Project → New Directory → New Project
 - a. Give your project a name and select an empty folder somewhere on your computer
2. Paste in the model files to the new directory (The three .R files and the PBKinput CSV file):

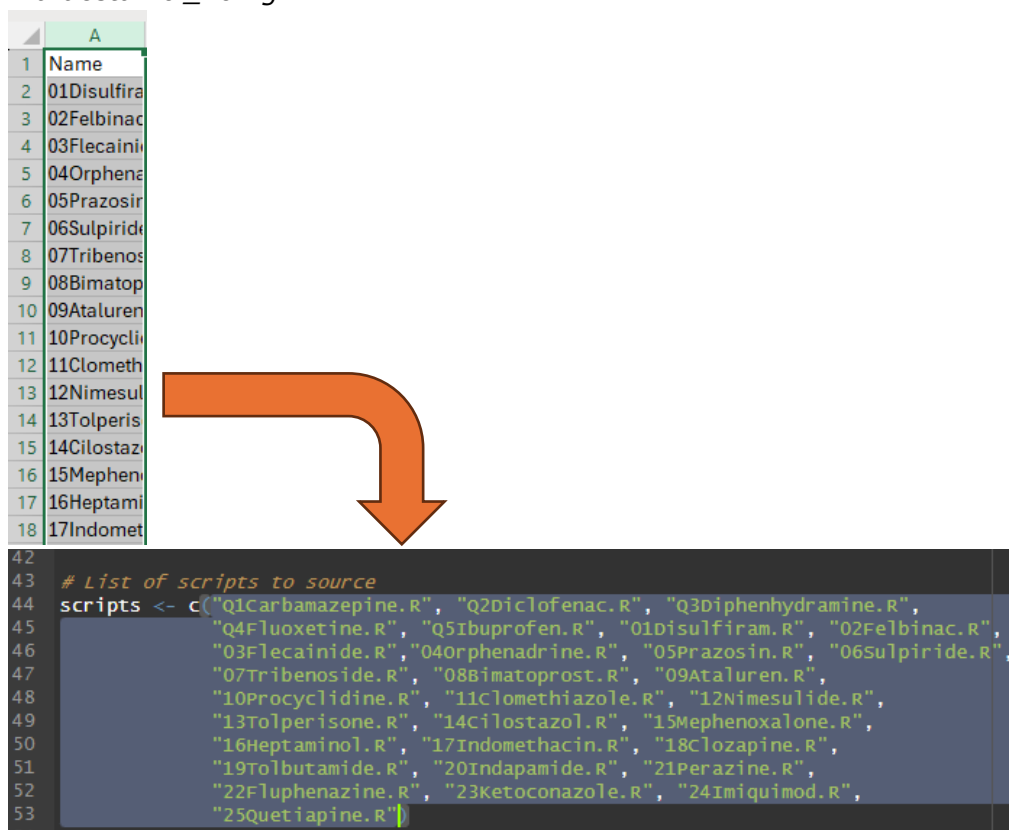


3. These should now appear in the file explorer in R Studio:

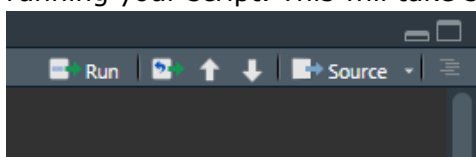


4. Open PBKinput.csv and insert your data. See above for details on each column

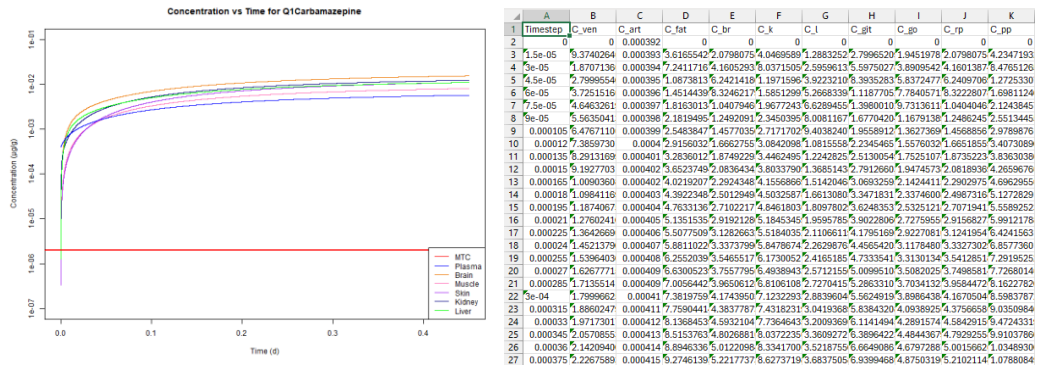
5. Open Parallel Batch Code.R in R studio. This will open the script. Scroll down to find where the code is going to use the names from the CSV, should be line 39.
 - a. Change the names in green to match the names you put into your CSV file. For example, if you have a row in your Input CSV named "Paracetamol_10mg", then the line in the code should read "Paracetamol_10mg.R":



6. Press the 'Source' button on the top right of the Batch Code.R script to begin running your script. This will take some time:



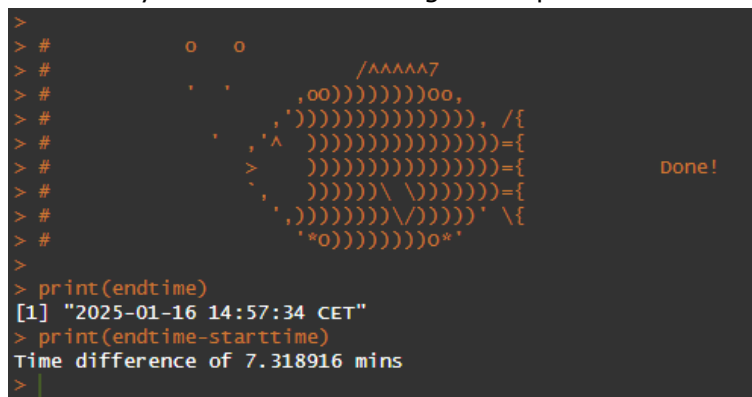
7. The model will now go through each compound one by one and generate a PBKOutputCSV and a graph. You can find both of these either in the folder, or in



- a. The Output CSV contains all of the compartments in the model, and the concentration at each timestep (measured in days). The length of the timesteps depends on the half-life of the compound (timestep = $HL/10,000$).

8. Done!

Once the model has finished running, an ASCII picture of a happy fish will appear in the console, along with the time difference between the start and end of the run, so that you know how long this process takes roughly for future runs.



a. The code also creates an excel file with a combination of all the final blood plasma results. You may also use this if it is useful to you.

Final Tips

- Intrinsic clearance has been added in the final update, you can enter this in the input CSV if you have that data. The model will run just fine without it if not.
- Timestep (dt) is quite important. If you get strange oscillations, or start to see infinities in your environment, then decrease the timestep by increasing the denominator in either of these equations:

```

39
40 #Amount of time between timesteps (days)
41 #If HL is used, use fraction of HL; if cl_int is used, use 1 second
42
43 if (!is.null(cl_int) && cl_int != 0) {
44     dt = (1/30000)/cl_int
45
46 }else{
47     dt = HL/20000 #Increase this number if you get oscillations
48 }

```

- You can change the definition of steady-state in the model file by changing the tolerance:

```

386 # Running the loop until steady state
387 i <- 1
388 max_iter <- 100000 # Prevent infinite loop
389 C_art_start = (Kx * (C_water - (C_ven * UF)/P_bw))/F_card #Used to find the steady state
390 # Define how close to zero is "steady state" based on starting arterial concentrations
391 # 0.01% of starting plasma concentration
392 tolerance <- (C_art_start/10000)
393

```